Advanced Optical Fiber Communications

Problem 4: Group velocity and dispersion

a) The energy velocity of an electromagnetic wave is defined as the ratio between the Poynting vector and the energy density. Show with the help of Maxwell's equations that the energy velocity is identical to the group velocity for a linearly x-polarized electromagnetic plane wave propagating in a dispersive material.

A plane electromagnetic wave is propagating inside silicon dioxide (SiO_2), which is a dispersive material.

b) Calculate the refractive index of SiO_2 at a wavelength of 1.55 µm with the Sellmeier equation.

c) Find the group index of SiO_2 and calculate the group velocity of the electromagnetic wave propagating through silicon dioxide. What is the dispersion coefficient D_M?

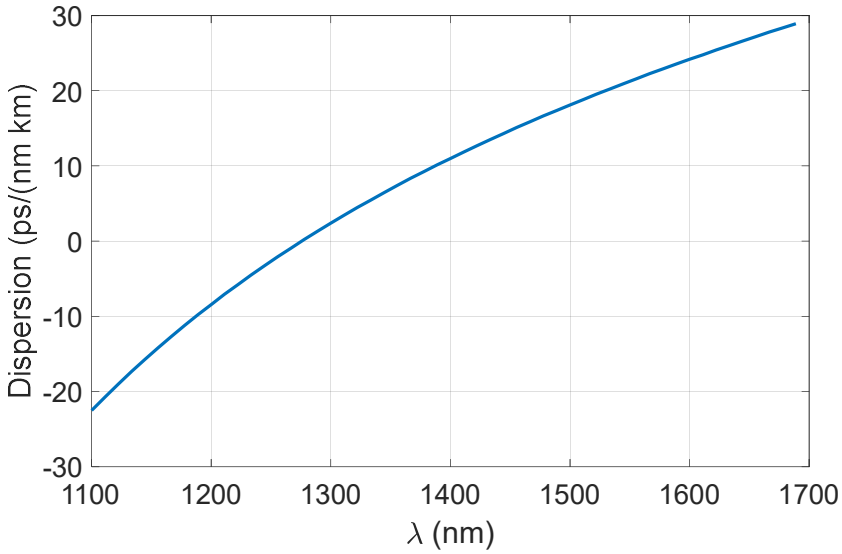


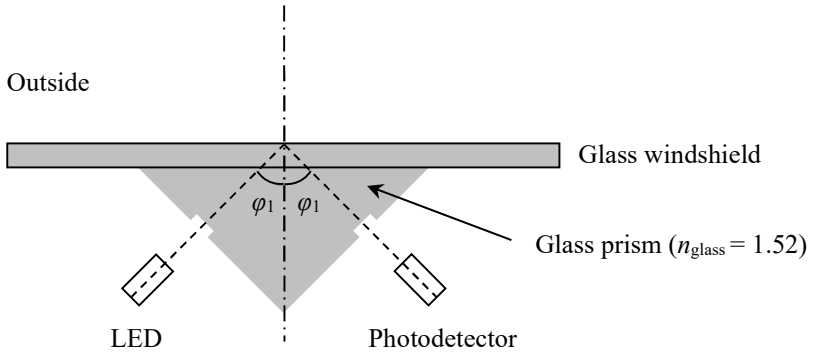
Figure 1 Plot of the dispersion coefficient of SiO_2

Problem 5: Attenuation

An electromagnetic wave propagates in a waveguide, e.g. an optical fiber, with a loss of 0.4 dB/km. Find the distances that the wave has to travel until the power decays to 1/2 (1/10, 1/100) of its initial value.

Problem 6: Rain sensor

A simplified graph of a rain sensor is illustrated in the figure below. The rain sensor is fixed on the glass windshield of a car ($n_{glass} = 1.52$). The LED and the photodetector are positioned so that $\varphi_1 = 45°$.



a) Calculate the critical angle for the case of a dry windshield and sketch the ray trace of the light from the LED notch to the photodetector notch. What is the power reflection coefficient for the perpendicular and parallel polarization?

b) Sketch the ray trace in the case of a wet windshield on a rainy day ($n_w = 1.33$). Calculate the power reflection coefficients of both polarizations.

c) What is the further procedure to improve the sensitivity of the rain sensor?

d) What would be the incident angle of the LED light, if no parallel polarization has to be detected by the photodetector on a rainy day?